

Load Test a Managed Instance Group and Escalate

120 minutes · Mapped criteria A5, A6, K2

Purpose and work product

Apply health checking and CPU autoscaling to a managed instance group, run a controlled load test and escalate a scaling failure.

Submit: load-results.csv, MIG configuration and escalation report

Safety, cost and privacy

Use only the assigned training project. Confirm the active account, project and region before every mutation. Never place credentials, payment details, access tokens or private keys in screenshots or submitted files. Stop billable resources and verify the final inventory.

Files in this folder

File	Purpose
load_test.sh	Mock data, script or configuration used by this activity
load-results.csv	Mock data, script or configuration used by this activity
escalation-template.md	Mock data, script or configuration used by this activity

Detailed procedure

Step 1

Use an approved managed instance group sandbox or create a small regional MIG from the Activity 4 template.

Step 2

Attach a health check whose path returns 200 without depending on a slow external service.

Step 3

Configure minimum 2, maximum 10, target CPU 60% and cooldown 90 seconds.

Step 4

Record the healthy baseline: group size, backend health, p95 latency and error rate.

Step 5

Run load_test.sh against the load-balancer URL and monitor group size plus CPU in parallel.

Step 6

Hold the load long enough to exceed the target beyond the cooldown. Record actual scale-out time.

Step 7

If scaling fails, inspect autoscaler status, quota, managed operations, health check and instance template.

Step 8

Apply one safe correction and repeat the identical load profile.

Step 9

Allow the group to recover and record scale-in behaviour.

Step 10

Complete load-results.csv with observed values and attach MIG/autoscaler JSON.

Step 11

Use escalation-template.md if the target still fails; identify the next owner and preserve evidence.

Step 12

Delete the load balancer and MIG or set the trainer-approved retained state. Verify final cost inventory.

Verification record

Record expected and observed output. A successful command without the correct project, identity, scope or evidence is not an accepted result.

Check	Expected evidence
Identity and scope	Active account, project ID, region or zone
Configuration	Resource description or policy output
Runtime result	Test output with UTC timestamp
Failure test	One controlled negative test and diagnosis
Cleanup	Final inventory and stopped/deleted billable resources

Troubleshooting sequence

Check the active identity and project first, then API enablement and IAM, then resource state, networking and quotas. Preserve the failed output before making a correction. If the issue remains, escalate with the exact command, timestamp, expected result, observed result and evidence files.

Cleanup

Delete or stop resources as instructed. Re-open Billing and the relevant service inventory, then record the remaining resource names. Budget alerts do not stop charges.